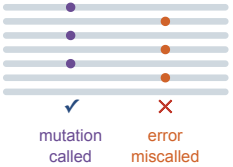
A Where the background comes from

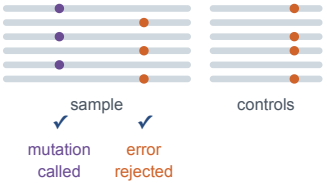
Quality-based test (e.g. LoFreq, Geneious)



Quality scores alone may not capture recurrent errors. High-quality errors from PCR, homopolymers or paralog reads can then be counted as mutations. Matched controls measure this background.

✓ correct decision ✕ wrong decision

Control-calibrated test (SHMtrace)



Error rates per substitution, strand and base quality are measured in matched AID-null libraries from the same run. A call needs support beyond this background, at an estimated FDR ≤ 0.05. This is the route for reads; a Sanger clone is counted instead (C).

B Inputs and read preparation



Nanopore
full-length amplicons



Sanger
.ab1 traces + sample sheet



Illumina
paired reads, merged



UMI-tagged
one consensus per molecule

Read preparation

1. **Length and expected-error gates**
   over-length reads and reads averaging > 0.01 errors per base
2. **Competitive alignment**
   minimap2 against target and paralog references; chimeras out
3. **Paralog-allele filter**
   reads with ≥ 10 paralog-specific bases removed
4. **Matched depth**
   random draw of the same number of usable reads per library

Sanger: traces are trimmed at quality ≥ 20 in place of step 1, and every clone is kept, so step 4 does not apply.

C Calling: a test for reads, a count for Sanger clones

Read libraries — nanopore, Illumina, UMI consensus: test candidate support

1 Control error model
per substitution, strand and base quality; a margin chosen by leave-one-control-out

2 Test and FDR
support against expected errors; estimated false fraction ≤ 0.05; held-out controls as decoys

Sanger traces — one molecule read once

1 Count the mismatches
every substitution at base quality ≥ 20 is counted; no per-call significance test in this route

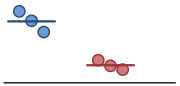
2 Background reported
AID-null clones give the calls the error alone would make, printed beside them and never removed

3 Germline and window
alleles at ≥ 50% removed; calls restricted to the covered window; indels optional

4 Automatic checks
spectrum (C→T/G→A share), positional clustering, 5-mer targeting index, Sanger bookkeeping flags

D Outputs

Animal-level tests and power



NB, Welch, permutation

Single-read estimate



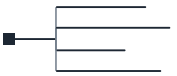
observed minus control background

Clones from the V(D)J junction



reads grouped by junction; each clone counted once

Mutation trees



full-length reads of one clone

Spectrum and motif shares



G:C / A:T, Ts / Tv, WRC/GYW, WA/TW

Reports, app and command line



HTML report, tables, project files